Alireza Sadeghi

(Updated: Apr. 2026)

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Education

- **Ph.D., Electrical Engineering, Clemson University**, Clemson, SC, USA — 2025-Present
- **M.Sc., Mechatronics Engineering, University of Tehran**, Tehran, Iran — 2018-2021
  (Top Iranian University by QS), GPA: 17.59 (out of 20)
  Thesis Title: Clinical Diagnosis Identification from 12-Lead ECG Using Deep Learning
- **B.Sc., Mechanical Engineering – Manufacturing and Production, Amirkabir University of Technology (Tehran Polytechnic),** Tehran, Iran — 2013-2018
  (3rd Top Iranian University by QS), GPA: 15.01 (out of 20)
  Thesis Title: Design, Analysis and Fabrication of Rehabilitation Soft-Robots

Experience

- Graduate Research Assistant at Clemson University — 2025-Present
- Research Assistant at Sari Medical Center — 2022-2025
- Teaching Assistant of Machine Learning Course, Dr. A. Rezaee — 2021
- Teaching Mathematics and Physics to High School Students — 2013 – 2017

Research Interests

AI for Healthcare	Deep & Machine Learning	Medical Signal & Image Processing
Explainable AI	AI Inference & Reasoning	Causal Deep Learning

Skills

Programming Languages:	Python (esp. PyTorch, TensorFlow, Keras, NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn), R, MATLAB, Fortran
Web Languages:	CSS, HTML
Software:	WEKA, LaTeX, Siemens PLC, CATIA, Microsoft Office

English & GRE

TOEFL: 1st September 2024	Overall: 108, Reading: 29, Listening: 30, Speaking: 23, Writing: 26
GRE: 21st August 2023	QR: 167; VR: 149; AW: 3.5

Publications

TITLE	CITED BY	YEAR
Causal Representation Learning on High-Dimensional Data: Benchmarks, Reproducibility, and Evaluation Metrics **A Sadeghi**, W AbdAlmageed arXiv (Submitted to Transactions on Machine Learning Research)		2026
Interpretable graph-based models on multimodal biomedical data integration: A technical review and benchmarking **A Sadeghi**, F Hajati, A Argha, NH Lovell, M Yang, H Alinejad-Rokny arXiv (Accepted by Nature Communications)	2	2025
3DECG-Net: ECG fusion network for multi-label cardiac arrhythmia detection **A Sadeghi**, F Hajati, A Rezaee, M Sadeghi, A Argha, H Alinejad-Rokny Computers in Biology and Medicine	23	2024
A deep learning-based model for detecting Leishmania amastigotes in microscopic slides: a new approach to telemedicine **A Sadeghi**, M Sadeghi, M Fakhar, Z Zakariaei, M Sadeghi, R Bastani BMC Infectious Diseases	23	2024
Potential diagnostic application of a novel deep learning-based approach for COVID-19 **A Sadeghi**, M Sadeghi, A Sharifpour, M Fakhar, Z Zakariaei, M Sadeghi, M Rokni, A Zakariaei, E S Banimostafavi, F Hajati Scientific Reports	27	2024
Scoping review of deep learning techniques for diagnosis, drug discovery, and vaccine development in Leishmaniasis **A Sadeghi**, M Sadeghi, M Fakhar, Z Zakariaei, M Sadeghi Transboundary and Emerging Diseases	6	2024
The clinico-epidemiological, diagnostic and therapeutic aspects of methanol poisoning: A five-year retrospective study, northern Iran M Sadeghi, M Fakhar, SM Hoseininejad, Z Zakariaei, **A Sadeghi** Drug and alcohol dependence	11	2023
Deep Learning Techniques on Cardiotocography Data for Detection of Fetal Distress: A Technical Review **A Sadeghi**, M Sadeghi, K Yaghoubinia, V Smith, R Galinsky, F Marzbanrad (Under the first revision in BMC Pediatrics)		
A Causal Multimodal Generative Model for Disentangled Representation Learning and Counterfactual Reasoning **A Sadeghi**, W AbdAlmageed Submitted to ECCV 2026 Conference		
GranTVAE: A Granger-based Transformer Variational Autoencoder for Multi-Channel Time Series Prediction **A Sadeghi** (In Progress)		

Academic Projects

GranTVAE: A Granger-based Transformer Variational Autoencoder for Multi-Channel Time Series Prediction

- Developed a novel framework integrating Granger causality into a Transformer-based VAE for multi-channel time series forecasting, enabling joint causal discovery and prediction.
- Modeled both inter-variable and intra-variable dependencies, improving robustness to noise, irregular sampling, and missing data.
- Achieved state-of-the-art performance in causal graph recovery and predictive accuracy, with ablation studies confirming the importance of the causal module.

A Causal Multimodal Generative Model for Disentangled Representation Learning and Counterfactual Inference

- Developed a multimodal causal representation learning framework combining image data with VLM-generated textual descriptions to enhance latent factor discovery.
- Designed a structured generative model that jointly learns disentangled representations, infers causal relationships, and enables counterfactual image generation.
- Demonstrated improved disentanglement and downstream task performance on synthetic and real-world benchmarks, outperforming unimodal baseline methods.

Potential diagnostic application of a novel deep learning- based approach for COVID-19

- Developed deep learning models for automated COVID-19 diagnosis from CT scans, including transfer learning approaches for slice-level analysis and a novel 3D MASERes architecture for patient-level prediction.
- Designed MASERes (Multi-Head Attention Squeeze-and-Excitation Residual Network), integrating multi-head attention, channel-wise recalibration (SE blocks), and residual connections to effectively capture spatial and inter-slice dependencies in 3D CT volumes.

3DECG-Net: ECG fusion network for multi-label cardiac arrhythmia detection

- Developed 3DECG-Net, a deep learning model for multi-label classification of seven cardiac arrhythmias, transforming 12-lead ECG signals into 3D representations using the Recurrent Plot technique.
- Designed a Conv-LSTM residual network with multi-head attention, effectively capturing both spatial and temporal dependencies across ECG leads while enabling feature fusion and robust representation learning.
- Incorporated explainable AI (LIME) analysis to interpret lead contributions, ensuring trustworthy and clinically relevant arrhythmia detection, alongside a preprocessing pipeline for high-quality ECG data.

A deep learning-based model for detecting Leishmania amastigotes in microscopic slides: a new approach to telemedicine

- Developed LeishFuNet, a deep learning framework for automated detection of Leishmania parasites in microscopic images, leveraging same-domain transfer learning from models pre-trained on COVID-19 datasets.
- Utilized a fusion of multiple pre-trained architectures (VGG19, ResNet50, MobileNetV2, DenseNet169) to enhance detection accuracy and robustness on high-resolution microscopic images.
- Incorporated Grad-CAM explainable AI to visualize and interpret model predictions, supporting trustworthy and clinically relevant telemedicine diagnostics.

Awards & Honors

- Recipient of Fully Funded PhD Scholarship, Clemson University, Clemson, SC, USA — 2025
- Ranked Within the Top 3% of Participants in the Data Science Pathway of CodeCup Competition (The biggest programming competition in Iran) — 2022
- Ranked Within the Top 4% Of Participants in The Nationwide Machine Learning & Data Science Competition Held by Amirkabir University of Technology — 2022
- Ranked Within the Top 10% Of GPAs Among the Students Who Graduated in 2021 from University of Tehran in the Mechatronic Engineering Discipline — 2021
- Ranked Within the Top 14% of Participants in the Kaggle Competition: RSNA-MICCAI Brain Tumor Radiogenomic Classification — 2021
- Ranked Within the Top 1% in the Iranian University Entrance Exam for Master of Mechatronic Engineering *(2018)* — 2018
- Ranked Within the Top 0.5% in the Mathematics and Physics Iranian University Entrance Exam — 2013
- Admitted in the First Step of National Olympiads of Astronomy and Physics — 2011
- Admitted In the First Step of National Olympiad of Astronomy — 2010
- Ranked 2nd in the National Competition of "NODET Projects" (Which is Held Among NODET Students) — 2007
- Admitted to the NODET Junior High School (National Organization for Development of Exceptional Talents) — 2006

References

Dr. Farshid Hajati (M.Sc. Supervisor)
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Dr. Hamid Alinejad-Rokny (Paper Co-author)
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